

Skills Summary

Equations with Variables on Both Sides

Use this reference while completing your worksheet or when studying.

Skill 1 — Collecting the variable terms

The balance rule: whatever is done to one side must be done to the other.

Order of moves: clear brackets, take the *smaller* variable term off both sides, undo the constant, undo the coefficient. Taking the smaller variable term off keeps the surviving coefficient positive, which is why that choice is worth making.

Every move uses the **inverse** operation: a subtraction is undone by adding.

Example: Solve $5x + 2 = 3x + 10$.

Step 1. Variables sit on both sides, so nothing can be undone until one side is variable-free — remove the smaller variable term from both sides first.

Step 2. That leaves a two-step equation: subtract the constant from both sides, then divide both sides by the surviving coefficient.

$$\Rightarrow x = 4$$

Try it: Solve $6x - 1 = 2x + 11$.

check: $x = x$

Skill 2 — Building the equation from a situation

Method: name the unknown with a letter and say what it counts. Write an expression for each side — a fixed amount plus a per-unit amount times the unknown — and set them equal, because "the same" is what an equals sign means.

Check the answer in the *story*, not only in the equation: an answer must make sense as a count of classes, hours, or metres.

Example: One fair charges 6 to enter plus 2 a ride; another charges 2 to enter plus 3 a ride. After how many rides is the cost the same?

Step 1. The question asks when two totals are equal, so the unknown is the number of rides and the equation sets the two cost expressions equal.

Step 2. Each side is an entry charge plus a rate times the rides; taking the smaller rides term off both sides leaves a one-step equation.

$$\Rightarrow r = 4 \text{ rides}$$

Try it: One pool charges 10 plus 2 a swim; another charges 4 plus 4 a swim. After how many swims is the cost equal?

check: $x = x$

Skill 3 — When the variable disappears

Two special cases. Collecting the variable terms sometimes removes them completely, and what is left decides everything:

True statement left (such as $6 = 6$): every number works — infinitely many solutions.

False statement left (such as $1 = -3$): no number works — no solution.

Example: Solve $3(x + 2) = 3x + 6$, then $4x + 1 = 4x - 3$.

Step 1. Both have matching variable terms, so I clear brackets and collect, then read whatever numerical statement survives.

Step 2. The first distributes to give the two sides identical; the second loses its variable terms and leaves two different numbers claimed equal.

First: $3x + 6 = 3x + 6$, true for every x . Second: no solution.

Try it: How many solutions has $5(x - 2) = 5x - 10$?

check: $5x - 10 = 5x - 10$, infinitely many